

Predicting Voter Turnout with tidymodels

A Modern Machine Learning Ecosystem in R

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Who votes — and can we predict it?

- ~60% of eligible Americans voted in 2020, but turnout varies a lot by county
- Campaigns and policymakers want to know: who shows up?
- We have covered regression and inference — tidymodels asks a different question: how accurately can we *classify* counties?

From inference to prediction

What we have done

- Regression: why does turnout vary?
- Inference: is income significant?
- Bootstrapping: how certain are we?

What tidymodels adds

- How accurately can we classify?
- Honest train/test evaluation
- One workflow for any model

`tidymodels` is to machine learning what `tidyverse` is to data wrangling.

What is tidymodels?

A family of packages, one grammar for the whole modeling process:

Package	Role
<code>rsample</code>	Train/test splits
<code>recipes</code>	Preprocessing
<code>parsnip</code>	Model specification
<code>workflows</code>	Bundle recipe + model
<code>yardstick</code>	Evaluate performance

Same tidy principles — just applied to modeling.

The tidymodels workflow



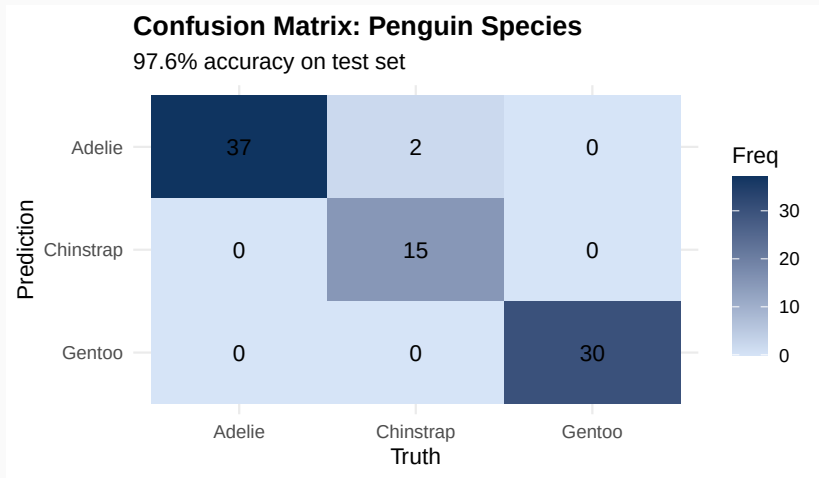
Each step is modular — swap the model without rewriting anything else.

Vignette: penguin species classification

Official tidymodels tutorial — predicting species from physical measurements

```
splits <- initial_split(penguins, prop = 0.75,  
                        strata = species)  
train  <- training(splits)  
test   <- testing(splits)  
  
rec <- recipe(species ~ bill_length_mm +  
              bill_depth_mm + flipper_length_mm +  
              body_mass_g, data = train) %>%  
              step_normalize(all_numeric_predictors())  
  
wf <- workflow() %>%  
      add_recipe(rec) %>%  
      add_model(multinom_reg()) %>%  
              set_engine("nnet")) %>%  
      fit(data = train)
```

Vignette results — 97.6% accuracy



My application: voter turnout

Can county-level socioeconomic data predict whether a county has high or low turnout?

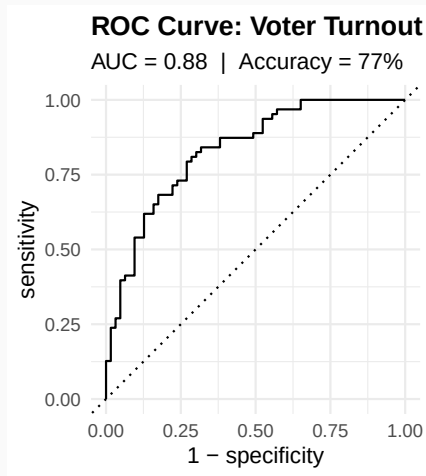
Predictor	Description
<code>median_income</code>	County median household income
<code>pct_college</code>	% adults with college degree
<code>pop_density</code>	Population per sq. mile
<code>pct_white</code>	Racial composition proxy
<code>region</code>	US Census region

Outcome: `high_turnout` — above/below county median (binary)

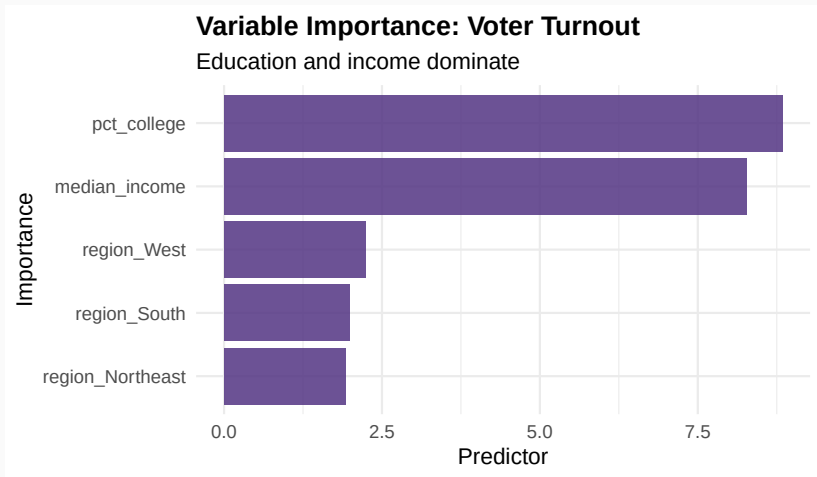
Building the model

```
splits <- initial_split(turnout, prop = 0.75,  
                        strata = high_turnout)  
  
rec <- recipe(  
  high_turnout ~ median_income + pct_college +  
    pop_density + pct_white + region,  
  data = train  
) %>%  
  step_normalize(all_numeric_predictors()) %>%  
  step_dummy(all_nominal_predictors())  
  
wf <- workflow() %>%  
  add_recipe(rec) %>%  
  add_model(logistic_reg() %>%  
    set_engine("glm")) %>%  
  fit(data = train)
```

Results: 77% accuracy, AUC 0.88



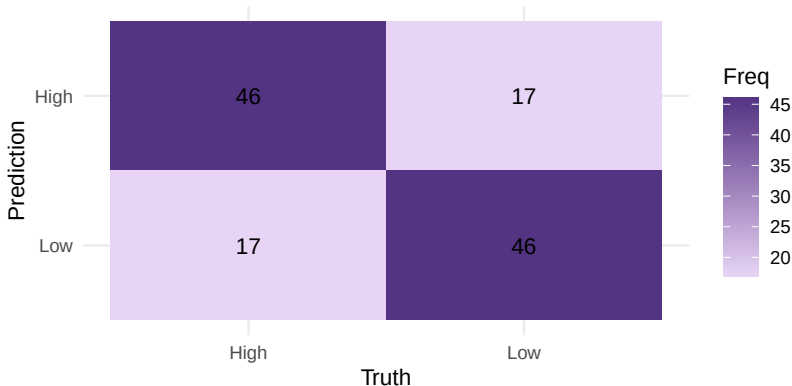
What predicts turnout?



Confusion matrix

Confusion Matrix: Voter Turnout

How often did the model get it right?



Takeaways

What we found

- Education and income are the strongest predictors — consistent with the political science literature

Why tidymodels?

- One workflow for any model — swap `logistic_reg()` for `rand_forest()` in one line
- Train/test split gives an honest accuracy estimate
- Natural extension of regression and inference from class

Limitations

- Ecological fallacy: county averages are not individual behavior
- Simulated data — real application would use MIT Election Lab

Thank you

- tidymodels documentation: tidymodels.org
 - MIT Election Lab: dataverse.harvard.edu
 - GitHub: github.com/your-username/tidymodels-voter-turnout
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